

Time Value of Money (TVM) – Lesson Notes

1. What is the Time Value of Money?

The Time Value of Money (TVM) is the idea that money available today is worth more than the same amount of money in the future. This is because money today can be invested to earn a return.

2. Key Concepts

Concept	Explanation
Present Value (PV)	The value of money today.
Future Value (FV)	The value of money at a future date after earning interest.
Interest Rate (r)	The rate of return earned per period.
Number of Periods (t)	The number of time periods money is invested.
Compounding	Interest earned on both the original amount and accumulated interest.

3. Future Value Formula

$$FV = PV \times (1 + r)^t$$

This formula calculates how much an investment today will grow to after earning interest for a certain number of periods.

4. Present Value Formula

$$PV = FV / (1 + r)^t$$

This formula calculates how much a future payment is worth today after discounting for interest.

5. Applications of TVM

TVM is widely used in finance for investment valuation, loan calculations, retirement planning, and corporate finance decisions such as discounted cash flow analysis.